

Geriatrics: Safe Medication Use in Older Adults

Introduction

- Aging population growth and increasing lifespan demand healthcare professionals adapt to older patients' needs.
- Drug-related problems are common in older patients, causing additional morbidities despite available pharmacotherapies.
- Clinicians should understand key aspects of geriatric care, including:
 - Epidemiology of aging.
 - Age-related physiological changes.
 - Prevalence of drug-related problems.
 - Comprehensive geriatric assessment.
 - Interprofessional approaches to care.

Age-Related Changes

- **Pharmacokinetics vs. Pharmacodynamics**
 - Pharmacokinetics: What the body does to the drug.
 - Pharmacodynamics: What the drug does to the body.
- **Aging effects on pharmacokinetics:**
 - All components (absorption, distribution, metabolism, excretion) are impacted.
 - Most clinically significant: Reduced renal elimination.
- **Frailty and variability in older patients:**
 - Changes influenced by clinical state, not just age.
 - Older adults more likely to be malnourished or have comorbidities affecting drug metabolism.
- **Clinical responsibilities for safe medication use:**
 - Account for altered pharmacokinetics and pharmacodynamics during prescribing and monitoring.
 - Start certain chronic medications (e.g., antihypertensives, antidepressants) at 50% of the standard initial adult dose.

- Gradual dose titration recommended, except for acute illness treatments (e.g., antibiotics for pneumonia).

Pharmacokinetic Changes

Absorption

- **Gastrointestinal (GI) changes with aging:**
 - Decreased intestinal epithelial surface, gastric acid secretion, and splanchnic blood flow.
 - Weaker peristalsis and delayed gastric emptying.
 - Slower absorption of enteric-coated and delayed-release preparations.
 - Longer time to achieve peak drug effects but no significant alteration in total drug absorption.
- **Nutrient absorption:**
 - Achlorhydria reduces absorption of vitamin B12, calcium, and iron.
- **Skin changes:**
 - Epidermal and dermal atrophy reduce skin barrier function.
 - Decreased tissue perfusion leads to variable transdermal, subcutaneous, and intramuscular absorption.
 - Intramuscular injections should generally be avoided.
- **Buccal absorption:**
 - Reduced saliva production slows absorption through buccal mucosa.
 - Most drugs show clinically insignificant changes in absorption.

Distribution

- **Body composition changes:**
 - Lean body mass decreases (12%-19%) → Increased blood levels of muscle-distributed drugs (e.g., digoxin).
 - Adipose tissue increases (18%-36% in men; 33%-45% in women) →
 - Fat-soluble drugs (e.g., diazepam, amitriptyline, amiodarone) show increased volume of distribution (Vd).

- Results in higher tissue concentrations, prolonged action, increased half-life, and delayed steady-state serum concentrations.
- Total body water decreases (10%-15% by age 80) →
 - Lower Vd for water-soluble drugs (e.g., aspirin, digoxin, morphine).
 - Leads to higher plasma concentrations and increased toxicity risk.
- **Plasma albumin changes:**
 - Albumin concentration decreases (10%-20%) due to disease/malnutrition rather than age alone.
 - Acute illness causes rapid serum albumin decrease → Enhanced drug effects.
 - Highly protein-bound drugs (e.g., warfarin, phenytoin, diazepam):
 - Malnourished patients with hypoalbuminemia have a higher free drug percentage.
 - Risk of toxicity with drugs like phenytoin in hypoalbuminemic patients even at low total drug levels.
 - Careful interpretation of serum drug levels is necessary.

Metabolism

- **Factors Affecting Drug Metabolism**
 - Influenced by age, acute and chronic diseases, and drug–drug interactions.
 - Liver is the primary site of metabolism; aging reduces liver mass (20%-30%) and hepatic blood flow (up to 50%).
 - Reduced hepatic blood flow lowers drug delivery, metabolism, and increases half-life.
- **Effects on Specific Drugs**
 - Metabolic clearance decreases (20%-40%) for drugs like amiodarone, amitriptyline, and morphine.
 - Clearance unchanged for low hepatic extraction drugs.
 - Higher bioavailability seen in drugs with high extraction ratios (e.g., morphine, propranolol, levodopa, calcium channel blockers, tricyclic antidepressants, statins).
- **CYP450 System**

- Aging may affect liver enzymes (CYP450), leading to decreased oxidative phase I metabolism; evidence is inconsistent.
- CYP450 activity variations attributed to lifestyle (e.g., smoking), illness, or drug interactions.
- **Phase II Metabolism**
 - Aging does not affect drugs undergoing phase II metabolism (e.g., lorazepam, temazepam), though frailty may reduce conjugation.

Elimination

- **Renal Drug Elimination**
 - Most significant pharmacokinetic change in older adults.
 - Decline in renal blood flow, renal mass, glomerular filtration rate, and tubular secretion with age.
 - Average renal clearance declines by 50% from age 25 to 85 years, independent of disease.
 - Up to 33% of older adults may not experience age-related renal decline.
- **Clinical Implications**
 - Decreased renal clearance leads to prolonged drug half-life, higher serum drug levels, and increased ADR risk.
 - Special caution for narrow therapeutic index drugs (e.g., digoxin, aminoglycosides); dose adjustments and monitoring required.
- **Serum Creatinine Considerations**
 - Serum creatinine may remain normal despite reduced renal function due to lower muscle mass in frail patients.
 - Overlooking renal decline based solely on serum creatinine can lead to overdoses and ADRs.
- **Calculating Renal Function**
 - Use Cockcroft-Gault equation to estimate creatinine clearance and adjust drug doses.

When serum creatinine is expressed in mg/dL,

$$\text{Creatinine Clearance} = \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{\text{Serum creatinine (mg/dL)} \times 72} \times (0.85 \text{ if female})$$

When serum creatinine is expressed in $\mu\text{mol/L}$,

$$\text{Creatinine Clearance} = 1.2 \times \frac{(140 - \text{Age}) \times \text{Weight (kg)}}{\text{Serum creatinine } (\mu\text{mol/L})} \times (0.85 \text{ if female})$$

- Formula considerations:
 - Adjust for body weight in obese patients ($\text{AjbW} = \text{IBW} + 0.4 [\text{ABW} - \text{IBW}]$).
 - Predictive formulas may overestimate renal function in chronically ill older adults.

Pharmacodynamic Changes

- **Definition:** Actions of a drug at its target site and the body's response.
- **Sensitivity:** Increased sensitivity to drug effects in older adults due to:
 - Reduced homeostatic mechanisms.
 - Altered receptor properties (binding affinity, receptor density, structural changes, post-receptor effects).
- **Receptor changes:**
 - Involves adrenergic, cholinergic, dopaminergic systems, GABA, and opioid receptors.
 - Can lead to both increased sensitivity and decreased response to certain drugs.

Cardiovascular System

- **Susceptibility to orthostatic hypotension:**
 - Decreased arterial compliance and baroreceptor reflex response.
 - Affects 5%–33% of older adults.
- **Drugs causing orthostasis:**

- Antipsychotics, direct vasodilators, loop diuretics, opioids, and antihypertensives.
- **Decreased β -adrenergic receptor function:**
 - Reduced sensitivity to β -agonists and β -antagonists in cardiovascular and respiratory systems.
 - Unchanged response to α -agonists and antagonists.
- **Specific drug effects:**
 - Increased response to calcium channel blockers (e.g., verapamil).
 - Higher risk of QT prolongation and torsade de pointes.
- **Clinical recommendations:**
 - Start at low doses, titrate slowly, and monitor for adverse effects.

Central Nervous System (CNS)

- **Increased sensitivity to CNS drugs:**
 - Lower doses often adequate; standard or high doses may cause adverse effects (e.g., oversedation, respiratory depression).
 - Examples: opioids, benzodiazepines, antidepressants, antipsychotics, neuroleptics, antihistamines.
- **Blood–brain barrier changes:**
 - Becomes more permeable, allowing more drugs to cross and cause CNS effects.
- **Cholinergic system changes:**
 - Decreased cholinergic neurons and receptors (nicotinic/muscarinic).
 - Decreased choline uptake and increased acetylcholinesterase.
- **Neurotransmitter imbalances:**
 - Leads to memory/movement disorders.
- **Dopaminergic system changes:**
 - Increased dopamine type 2 receptors → greater susceptibility to delirium.
 - Reduced dopamine neurons → higher incidence of extrapyramidal symptoms from antidopaminergic drugs (e.g., antipsychotics).

Fluids and Electrolytes

- Fluid and electrolyte homeostasis is reduced in older adults.
- Older adults experience:
 - **More severe dehydration** with the same fluid loss compared to younger adults.
 - Increased incidences of:
 - Hyponatremia
 - Hyperkalemia
 - Prerenal azotemia (especially with diuretics like hydrochlorothiazide or furosemide).
- Contributing factors:
 - Decreased thirst and cardiovascular reflexes.
 - Reduced fluid intake.
 - Reduced kidney ability to concentrate urine.
 - Increased atrial natriuretic peptide levels.
 - Reduced aldosterone response to hyperkalemia.
 - Decreased antidiuretic hormone response.
- Medications with higher risks:
 - **ACE inhibitors** (e.g., increased risk of hyperkalemia and acute renal failure).
- Recommendations:
 - Start with **low doses** of high-risk medications.
 - Titrate slowly with **frequent renal function monitoring**.

Glucose Metabolism

- Aging is associated with:
 - Reduced glucose tolerance due to:
 - **Decreased insulin secretion.**
 - **Increased insulin resistance (lower sensitivity).**
 - Increased risk of hypoglycemia with sulfonylureas (e.g., glyburide, glipizide).
 - Resulting from impaired counter-regulatory responses to hypoglycemia.
- Impaired autonomic response:
 - Reduced ability to sense hypoglycemia symptoms (e.g., sweating, palpitations, tremors).

- Neurological symptoms of hypoglycemia may include:
 - Syncope
 - Ataxia
 - Confusion
 - Seizures

Drug-Related Problems

- Comorbidities and polypharmacy complicate older adults' health, particularly due to inappropriate medications.
 - **Statistics:**
 - 43.6% of emergency department visits leading to hospitalization in older adults are due to adverse drug events.
 - 59% of older Medicare beneficiaries have at least one medication-related problem.
 - Drug-related morbidity and mortality cost the US healthcare system \$528.4 billion in 2016.
 - Health outcomes affected: withdrawal effects, therapeutic failure, adverse drug events.
 - **Solution:** Collaboration among interprofessional providers and patients ensures appropriate therapy, reduces adverse drug events, and improves adherence and outcomes.
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Polypharmacy

- Defined as the concurrent use of multiple medications (criteria: ≥ 4 –10 medications).
- **Trends:**
 - 39% of older adults reported using ≥ 5 medications in 2012 (up from 24% in 1999).
 - 67% of older adults used polypharmacy in 2011 (up from 53% in 2006).
- Common contributors: dietary supplements, herbal products, and nonprescription medications.
- In nursing homes:

- 50.7% of patients with severe cognitive impairment used 5–9 medications.
 - 16.9% used ≥ 10 medications (excessive polypharmacy).
 - **Key Issues:**
 - Multiple providers treating comorbidities without coordinated care.
 - Medication reconciliation is crucial as the aging population grows.
 - **Recommendations:**
 - Evaluate all medications during each visit.
 - Discontinue medications without indication.
 - Support appropriate polypharmacy for multimorbid patients.
 - Perform comprehensive medication reviews to identify drug-related problems.
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Inappropriate Prescribing

- Definition: Prescribing medications with significant risk of adverse events when safer alternatives exist.
- Associated negative outcomes: confusion, falls, and mortality.
- Issues: Medications often continued after the initial indication has resolved.
- **Screening Tools:**
 - Beers criteria (2019): Lists 30 medications/classes categorized into five groups:
 - Inappropriate in most older adults.
 - To avoid in certain conditions.
 - To use with caution.
 - Drug–drug interactions.
 - Dose adjustments based on kidney function.
- **Examples of potentially inappropriate medications (Beers criteria):**
 - **Benzodiazepines** (e.g., diazepam, alprazolam): Cognitive impairment, falls.
 - **First-generation antihistamines** (e.g., diphenhydramine): Confusion, anticholinergic effects.
 - **Tricyclic antidepressants** (e.g., amitriptyline): Sedation, orthostatic hypotension.
 - **NSAIDs** (e.g., ibuprofen): GI bleeding, ulcers.

- **Strategies for improvement:**
 - Build partnerships with patients and caregivers.
 - Perform comprehensive medication assessments.
 - Screen for drug–drug and drug–disease interactions.
 - Use time-limited trials to assess benefits/risks of medications.
 - Trial discontinuation to reassess medication necessity.
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Undertreatment

- Underutilization of medications can lead to reduced functioning, increased morbidity, and mortality.
- Prevalence: Occurs across community, hospital, and long-term care settings.
- **Contributing factors:**
 - Multimorbidity and polypharmacy.
 - Cost, fear of nonadherence, or adverse effects.
 - Limited evidence in older adults.
 - Hesitation to increase doses or initiate secondary prevention therapies.
 - Ageism and liability concerns.
- **Management:**
 - Clinical assessment to balance benefits and harms of therapy.
 - Consider patient goals, prognosis, and therapeutic benefit timelines.
 - Avoid underprescribing by supporting adherence, addressing financial barriers, and improving clinical assessments.

Table 1–1 Common Categories of Geriatric Undertreatment

Therapy	Concern
Anticoagulation in patients with atrial fibrillation	Overly concerned with risk of bleeding or the risk of falls if anticoagulated
Malignant and nonmalignant pain	Hesitant to prescribe opioids due to possible cognitive and bowel side effects, concerns about addiction; patients may often be hesitant to use opioids

Antihypertensive therapy	Underestimate the benefit on stroke and cardiovascular event prevention and/or fail to add the second or third medication needed to attain control
β -Blocker treatment in heart failure	Concerned about complications in high-risk patients despite the substantial evidence of mortality benefit
Statin treatment for ASCVD	Underestimate benefit or have concerns about adverse events

ASCVD, atherosclerotic cardiovascular disease.

Adverse Drug Reaction (ADR):

- **Definition:** WHO defines ADR as a noxious, unintended reaction occurring at normal human doses for prophylaxis, diagnosis, or therapy.
- **Prevalence:**
 - Increases with polypharmacy.
 - Most common drug-related problem among older nursing home residents.
 - 25% of community-dwelling older adults (using ≥ 5 medications) experience ADRs.
 - 9% of hospitalizations in older adults are caused by ADRs.
- **High-Risk Medication Classes:**
 - Anticoagulants, antidiabetics, opioids.
- **Predictors of ADRs in Older Adults:**
 - Taking >4 medications.
 - Hospital stay >14 days.
 - 4 active medical problems.
 - Admission to general medical units (vs. geriatric wards).
 - History of alcohol use.
 - Lower Mini-Mental State Examination score (confusion, dementia).
 - Addition of 2–4 new medications during hospitalization.
- **Predictors of Severe ADRs:**
 - Certain medications (diuretics, NSAIDs, antiplatelets, digoxin).
 - Number of drugs taken.
 - Advanced age.
 - Comorbidities.

- **Medication-Related Falls:**

- Common and damaging ADR.
- Associated with premature institutionalization and early death.
- Risk factors include polypharmacy and drugs listed in Beers criteria.
- High-risk agents: benzodiazepines, sedative hypnotics, neuroleptics, antidepressants, antipsychotics.
- Prevention: Deprescribing via slow taper with close monitoring.

Nonadherence:

- **Consequences:**

- Escalates healthcare costs due to disease worsening and increased hospitalization.

- **Risk Factors in Older Adults:**

- Multimorbidities, cognitive deficits, polypharmacy, financial barriers.

- **Barriers to Adherence:**

- Lack of patient understanding.
- Provider's failure to educate.
- Complex regimens due to polypharmacy.
- Treatment of asymptomatic conditions (e.g., hypertension, dyslipidemia).
- Medication costs.

- **Interventions:**

- Older person-centered, multicomponent approaches:
 - Education.
 - Adherence aids.
 - Regular follow-ups.